I. Q1: Randomly Yours, 5 marks

A Gaussian random variable with mean 0 and variance 10 and an Exponentially distributed random variable with mean 1 fall in love at first chat. As they chat about meeting in person, they decide to do the following. Each one of them draws a number from their distribution independently of the other. In case both the drawn numbers have the same sign (positive or negative), they will meet in person. Else, they will not meet. Derive the probability that they will meet?

Let's call the Gaussian RV G
Let's call the Exponential RV E.

$$P[G \text{ meets } E] = P[G>0 \ \& \ E>0] + P[G<0 \ \& \ E<0]$$

$$= P[G>0]\,P[E>0] + P[G<0]\,P[E<0]$$

[$\because$ The random variables are independent]

E is Exp. Thus $P[E<0]=0$.

$$\therefore P[G \text{ meets } E] = P[G>0]\,P[E>0]$$

$$= P[G>0]\,(1). \quad [\text{Why??}]$$

$$= \tfrac{1}{2} \quad [\text{Why??}]$$

(Think what the Gaussian PDF with mean 0 looks like)

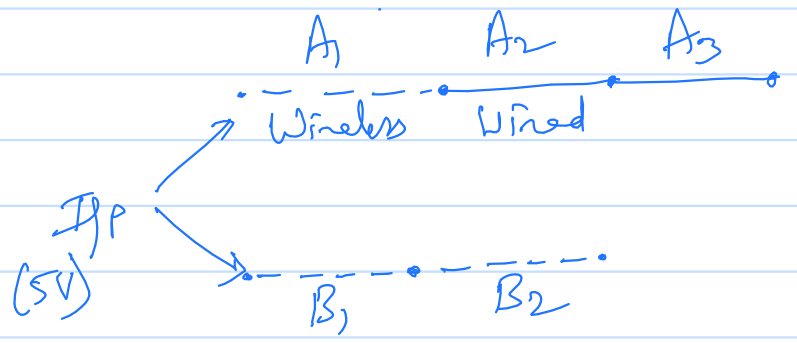
$\therefore$ 50% chance that G meets E!

(Q) What if both are Gaussians? What is the prob they meet?

(Q) What if both are exponentials? What is the prob they meet?

II. Q2: Path Diversity, 20 marks

You want to send a 5 V signal to your friend. You send the signal over two independent paths. Let's call them path A and path B. Path A consists of a wireless connection, followed by a wired connection, which is followed by another wired connection. On the other hand, path B consists of a wireless connection followed by another wireless connection. Each connection may or may not transmit a signal independently of the other. Suppose that a priori a wireless connection transmits the voltage with probability 0.5 and a wired connection transmits the voltage with probability 0.9. All connections in a path must transmit the voltage for your friend to receive the voltage along the path. Your friend receives 5 V as long as at least one of the paths transmits it. Else, your friend receives 0 V. Derive the expected value of the voltage received by your friend?



Let A_i indicate whether connection i of path A transmits a voltage. Similarly, define B_i.

Let $A_i = 1$ when i transmits a voltage
$A_i = 0$ when i doesn't transmit a voltage.

$P[A_1 = 1] = 0.5$ $\quad P[B_1 = 1] = 0.5$
$P[A_2 = 1] = 0.9$ $\quad P[B_2 = 1] = 0.5$.
$P[A_3 = 1] = 0.9$

Let R be the random variable that models the received signal.
Clearly the range space of R, $S_R = \{0, 5\}$

$$E[R] = 0\,P[R=0] + 5\,P[R=5]$$
$$= 5\,P[R=5]$$

$P[R=5] = P[$At least one path transmits the voltage$]$

$$= P[(A_1=1, A_2=1, A_3=1) \cup (B_1=1, B_2=1)]$$
$$= P[A_1=1, A_2=1, A_3=1] + P[B_1=1, B_2=1]$$
$$- P[A_1=1, A_2=1, A_3=1, B_1=1, B_2=1]$$
$$= (0.5)(0.9)^2 + (0.5)(0.5)$$
$$- (0.5)(0.9)^2(0.5)(0.5)$$

[We used that the connections behave independently of each other]

$$= (0.5)(0.9)^2(0.75) + 0.25$$
$$= (0.5)(0.9)^2(0.75) + (0.5)^2$$
$$= 0.5[0.56] + (0.5)^2 = (0.5)(1.06) \approx 0.53$$

$\therefore E[R] \approx 5(0.53)$ [Exact value of the prob is 0.55375]

[Of course, there are other approaches to the problem]

III. Q3: Ticket Sales, 20 marks

You have 2 movie tickets to sell within 4 minutes. During the first 2 minutes, you decide to price your tickets at ₹ 2. In the last 2 minutes you plan to sell any remaining tickets at ₹ 1. The number of buyers that approach you to buy tickets is a Poisson distributed random variable with rate 1/2 per minute. Any buyer buys exactly one ticket. Buyers arrive independently of each other. What is the expected value of the money you get as a result of your ticket sale? [Hint: The PMF of a Poisson RV K is $P[K=k] = \frac{(\lambda T)^k e^{-\lambda T}}{k!}$, $k = 0, 1, \ldots$, and $P[K=k] = 0$ otherwise.]

You may sell 0 or 1 or 2 tickets in the first two minutes

Event	Sale in First two minutes	Sale Next two minutes
E_{00}	0	0
E_{01}	0	1
E_{02}	0	2
E_{10}	1	0
E_{20}	2	Don't Care.
E_{11}	1	1

Note that the money you earn, say M, is a RV that maps the above events to numbers (rupees)

Event	Rupees earned
E_{00}	0
E_{01}	1
E_{02}	2
E_{10}	2
E_{20}	4
E_{11}	$2+1=3$

We want $E[M]$.

$P[M=0] = P[E_{00}] = P[$0 customers in first two minutes and 0 in the next two$]$

$= P[0 \text{ in first two}]\, P[0 \text{ in next two}]$

$= e^{-(1/2)2}\, e^{-(1/2)2}$

$= e^{-2}$

$P[M=1] = P[0 \text{ in first two}]\, P[1 \text{ in next two}]$

$= e^{-(1/2)2}\, \dfrac{(1/2 \cdot 2)\, e^{-(1/2)2}}{1!}$

$= e^{-2}$

$P[M=2] = P[E_{02} \cup E_{10}]$

$= P[E_{10}] + P[E_{02}]$

$= P[1 \text{ in first two } \& \text{ 0 in next two}] + P[0 \text{ in first two } \& \text{ At least two customers arrive in the next two}]$

$= \dfrac{(1/2)2\, e^{-(1/2)2}}{1!}\, e^{-(1/2)2} + e^{-(1/2)2}\left(1 - e^{-(1/2)2} - \dfrac{(1/2)2\, e^{-(1/2)2}}{1!}\right)$

$= e^{-2} + e^{-1}(1 - e^{-1} - e^{-1})$

$= e^{-2} + e^{-1} - e^{-2} - e^{-2} = e^{-1} - e^{-2}$

$P[M=4] = P[E_{20}]$

$= P[\text{At least 2 in the first two minutes}]$

$= 1 - e^{-1} - e^{-1}$

$= 1 - 2e^{-1}$

$P(M=3) = P[E_{11}]$

$= P[\text{One in first two}]\, P[\text{One in next two}]$

$= (e^{-1})(e^{-1})$

$= e^{-2}$

We didn't need to calculate $P[M=0]$!

$E[M] = 0\,P[M=0] + 1\,P[M=1] + 2\,P[M=2] + 3\,P[M=3] + 4\,P[M=4]$

$= e^{-2} + 2(e^{-1} - e^{-2}) + 3e^{-2} + 4(1 - 2e^{-1})$

$= e^{-2} + 2e^{-1} - 2e^{-2} + 3e^{-2} + 4 - 8e^{-1}$

$= 2e^{-2} + 2e^{-1} + 4 - 8e^{-1}$

$= 4 - 6e^{-1} + 2e^{-2}$ //

IV. Q4: Curious Coin, 20 marks

You create a coin that has the following properties. When the coin is tossed, the coin internally starts a timer that ends after a time interval that is exponentially distributed with rate 1 per microsecond. If the timer ends in 0.5 microseconds, then the coin toss gives heads. If the clock expires after 1.5 micro seconds, the coin toss gives tails. Otherwise, the coin keeps spinning (and no result is obtained). We will associate a value of 0.5 with heads, 1.5 with tails, and 1 with spinning. Answer the following questions. [Hints: Expectation of a sum of random variables is the sum of expectations of the random variables. The pdf of a rate λ exponential RV X is $f_X(x) = \lambda e^{-\lambda x}$, for $x \geq 0$, and 0 otherwise.]

1) Consider ten independent coin tosses of your coin. What is the expectation of the sum of values of all coin tosses?
2) Consider ten independent coin tosses of your coin. Suppose all coin tosses are known to have resulted in either heads or tails. What is the expectation of the sum of values of all coin tosses?
3) Consider ten independent coin tosses of your coin. Suppose that the 10^{th} coin toss is known to have resulted in spinning. There is no information about the other tosses. What is the expectation of the sum of values of all coin tosses?

H: Heads $\rightarrow$ 0.5
S: Spinning $\rightarrow$ 1
T: Tails $\rightarrow$ 1.5

Let X be the Exp RV.
λ is 1 per micro second.

$$P[H] = P[X \leq 0.5] = 1 - e^{-0.5}$$

$$P[T] = P[X > 1.5] = e^{-1.5}$$

$$P[S] = 1 - P[H] - P[T] = 1 - (1 - e^{-0.5} + e^{-1.5})$$
$$= e^{-0.5} - e^{-1.5}$$

Let Y_i denote the outcome of the i^{th} coin toss.

$$E[Y_i] = 0.5\,P[H] + 1\,P[S] + 1.5\,P[T]$$
$$= 0.5(1 - e^{-0.5}) + e^{-0.5} - e^{-1.5} + 1.5 e^{-1.5}$$
$$= 0.5 + 0.5e^{-0.5} + 0.5e^{-1.5}$$

This is for any coin toss i.

(1) Expectation of sum of values of 10 coin tosses is

$$10\,E[Y_i]$$

(2) We need to revise our beliefs about the $P[H]$, $P[T]$, and $P[S]$.
That is, we now need the conditional PMF of Y_i, conditioned on the knowledge that coin toss i either led to Heads or Tails. This is the same as conditioning on the knowledge that Spinning didn't result from any of the observed coin tosses.

$$P[Y_i = 0.5 \mid \text{Coin toss didn't result in a spin}]$$

$$= \frac{P[Y_i = 0.5, \text{Coin toss didn't result in a spin}]}{P[\text{Coin toss didn't result in a spin}]}$$

$$= \frac{P[Y_i = 0.5]}{P[H] + P[T]}$$

Note that $Y_i = 0.5$ is the same as the event Heads on coin toss i.
$H \cap (H \cup T) = H$

$$= \frac{P[H]}{P[H] + P[T]} = \frac{1 - e^{-0.5}}{1 - e^{-0.5} + e^{-1.5}}$$

$$P[Y_i = 1.5 \mid \text{Coin toss didn't result in spinning}]$$

$$= \frac{P[T]}{P[H] + P[T]} = \frac{e^{-1.5}}{1 - e^{-0.5} + e^{-1.5}}$$

$$\therefore E[Y_i \mid \text{Coin toss didn't result in spinning}]$$

$$= \frac{0.5\,P[H] + 1.5\,P[T]}{P[H] + P[T]}$$

10 such coin tosses $\Rightarrow$

$$10\left(\frac{0.5\,P[H] + 1.5\,P[T]}{P[H] + P[T]}\right)$$

(3). You are interested in outcomes that have a known outcome of spinning for the 10^{th} coin toss.
For each of the other 9 coin tosses $E[Y_i]$ is as we calculated for part (1).

The expected value is $9\,E[Y_i] + 1$

The expected value of the toss is simply the value of spinning.

V. Q5: Testing Communications, 35 marks

You are given the task of testing a wireless link between Delhi and Hyderabad. You do so by sending many bits (each bit is either 0 or 1) over the link from Delhi to Hyderabad. A receiver at Hyderabad records, for every sent bit, its estimate (0 or 1) of what was sent. At the end of the experiment you come up with the following data. You find that 90% of the bits sent are received correctly. You also find that amongst the correctly received bits, 70% of the bits were sent as 1. Also, 30% of the wrongly received bits were sent as 1.

The cost of sending a bit that is correctly received is -100 and that of sending a bit that is wrongly received is 50. What is the expected cost of sending a bit over the link?

[This question is as close to a freebie as one can be]

$P[\text{Bit is received correctly}] = 0.9$

$P[\text{Bit sent is } 1 \mid \text{Bit was received correctly}] = 0.7$.

$P[\text{Bit sent is } 1 \mid \text{Bit was wrongly rcvd}] = 0.3$

Cost C is a RV:

$$C = \begin{cases} -100 & \text{when bit is correctly rcvd,} \\ 50 & \text{when bit is wrongly rcvd.} \end{cases}$$

$P[C = -100] = P[\text{bit is correctly rcvd}]$

$= 0.9$.

$P[C = 50] = P[\text{bit is wrongly rcvd}] = 0.1$.

$E[C] = (-100)(0.9) + 50(0.1)$

$= -90 + 5 = -85$ //